

Revisiting *Remarks*: the learning of English nominals

1 Introduction. The central focus of Chomsky’s (1970) *Remarks on Nominalization* is the distinction between gerundive nominals, as in (1b), and derived nominals, as in (1c).

- (1) a. John refused the offer b. John’s refusing the offer c. John’s refusal of the offer

Subsequent work has proposed various theoretical mechanisms to account for this distinction, either via architectural principles (e.g. locating their formation in the syntax or lexicon, as in *Remarks*) or structural differences (e.g. Abney 1987, Marantz 1997, Alexiadou 2001, Wood 2023).

In Chomsky’s original formulation, the first and foremost difference between gerundive nominals and derived nominals is in their productivity: the former can be formed productively from VPs, whereas the latter cannot (Chomsky 1970: p. 16). This is the focus of the present work. I provide a learning-based explanation for how their difference in productivity arises.

2 Learning via the Tolerance Principle. I adopt the view of learning underlying the Tolerance Principle (TP; Yang 2016), where a child acquires a productive rule iff they receive sufficient positive evidence for it (Nida 1949, Aronoff 1976, Marchman 1988, Bybee 1995). The TP gives a concrete criterion for sufficiency: (2). Over the course of development, the child acquires rules that build on each other. The resulting system of rules is their grammar.

- (2) The Tolerance Principle

Let R be a rule applying to a domain of size N . Let M be the number of items in the domain following R that a learner encounters in their input. Then, the learner generalizes rule R in its domain if and only if $M > N - \frac{N}{\ln N}$.

Since children learn the core grammar at a young age, we only need to check the productivity of the rules on a small vocabulary. Specifically, *-ing* is productive by age 2, when the child’s vocabulary consists of at most 100 verbs (Brown 1973). I thus use the most frequent 100 verbs in my corpus as the child’s vocabulary of verbs. All counts and examples come from a 1.7 million-token corpus of child-directed speech (Brown 1973, MacWhinney 2000, Demuth et al. 2006).

3 Learning gerundive nominals. While the linguist takes “gerundive nominals” as the starting point of inquiry, the child has a different starting point: they are concerned with making sense of their input. Their input includes many phrases headed by words consisting of a verb with the suffix *-ing*; see (3). I call these phrases iP. The child’s task is to learn the rules that govern iPs, which must account for their internal structure and external distribution.

- (3) a. the horses are [running] d. all those fish eyes [looking at me]
b. you’re [playing candyland] e. I thought [reading books] was your favorite
c. you sure like [making that sword] f. the [going to bed] book

With respect to internal structure, the input supports a productive rule over VP: an iP is formed by taking any VP, then adding *-ing* to the verb. The productivity of this rule is due to two factors. First, the *-ing* suffix is productive in the domain of verbs, as $M = 98$ of the most frequent 100 verbs are attested in *-ing*, surpassing the threshold of 79 required for generalization by the TP. This establishes a productive rule over Vs. Second, iPs occur with internal material diagnostic of VPs, e.g. direct objects. Thus children can easily identify that iPs have the internal structure of VPs. In *Remarks*, this productive mapping between a VP and iP is represented as a transformational rule applying to VPs. We can also represent this as *-ing* attaching above VP (as in Abney 1987); concretely, [_{iP} *-ing* VP].

The child must then learn rules governing the external distribution of iPs. As (3) shows, iPs enter into various environments beyond gerundive nominals, which each must be learned. But relevant to us is the rule that iP can distribute like an NP; concretely, [_{NP} n iP]. For this to be productive, sufficiently many iPs must be attested in NP environments. Here, I take NP environments to be the complements of prepositions and verbs that take nominal objects, as in (4). Of the most frequent 100 verbs, $M = 84$ appear as an iP head in such an environment, which is sufficient for generalization. Thus iPs can be

(4) a. it's not for [playing] d. you like [going to her house]
 b. thank you for [closing the door] e. let's stop [throwing them]
 c. he's very serious about [eating these grapes]

4 Learning derived nominals. For derived nominals, in contrast, there is no evidence to motivate a productive rule applying to VP. First, unlike *-ing*, other nominalizing suffixes are not productive over V. For example, *-ment* and *-al* each only occur with one out of the most frequent 100 verbs (*movement* and *trial*), while *-ion* occurs with none. Instead of a productive rule over V, the child memorizes the nominalizing suffix for each verb. Note that this does not preclude the existence of additional restrictions imposed by nominalizers or productivity within subdomains (Aronoff 1976). Second, the child has no evidence to posit that the phrases headed by these derived nouns contain VP; for example, derived nouns never take nominal complements.

5 Discussion. I have now shown, as desired, how the difference in productivity between gerundive and derived nominals arises from what rules the input can support. For gerundive nominals, the input supports a productive rule from VP to iP, and a productive rule allowing iPs to occur in NP environments. For derived nominals, no analogous rules are supported.

My analysis contributes to a growing body of work that takes an acquisition-based approach to language (Kodner 2023, Belth 2024, Legate & Yang 2025, Lu et al. to appear). From this perspective, the contrast in *Remarks* can be understood as a consequence of learning, rather than architectural or structural principles.

References. Abney (1987), *The English noun phrase in its sentential aspects*. Alexiadou (2001), *Functional structure in nominals*. Aronoff (1976), *Word formation in generative grammar*. Belth (2024), “A learning-based account”, *Linguistic Inquiry*. Brown (1973), *A first language*. Chomsky (1970), “Remarks on nominalization”. Demuth, Culbertson, & Alter (2006), “Word-minimality”, *Language & Speech*. Grimshaw (1990), *Argument structure*. Kodner (2023), “What learning Latin”, *NLLT*. Legate & Yang (2025), “A null theory of movement”, manuscript. Lu, Legate, & Yang (to appear), “The learnability of bridge effects”, *Journal of Linguistics*. MacWhinney (2000), *The CHILDES project*. Marantz (1997), “No escape from syntax”, *U. of Penn. Working Papers in Linguistics*. Wood (2023), *Icelandic nominalizations and allosemy*. Yang (2016), *The price of linguistic productivity*.